

The logo for GA GUARDIAN, featuring a stylized 'GA' in a white serif font followed by the word 'GUARDIAN' in a white sans-serif font.

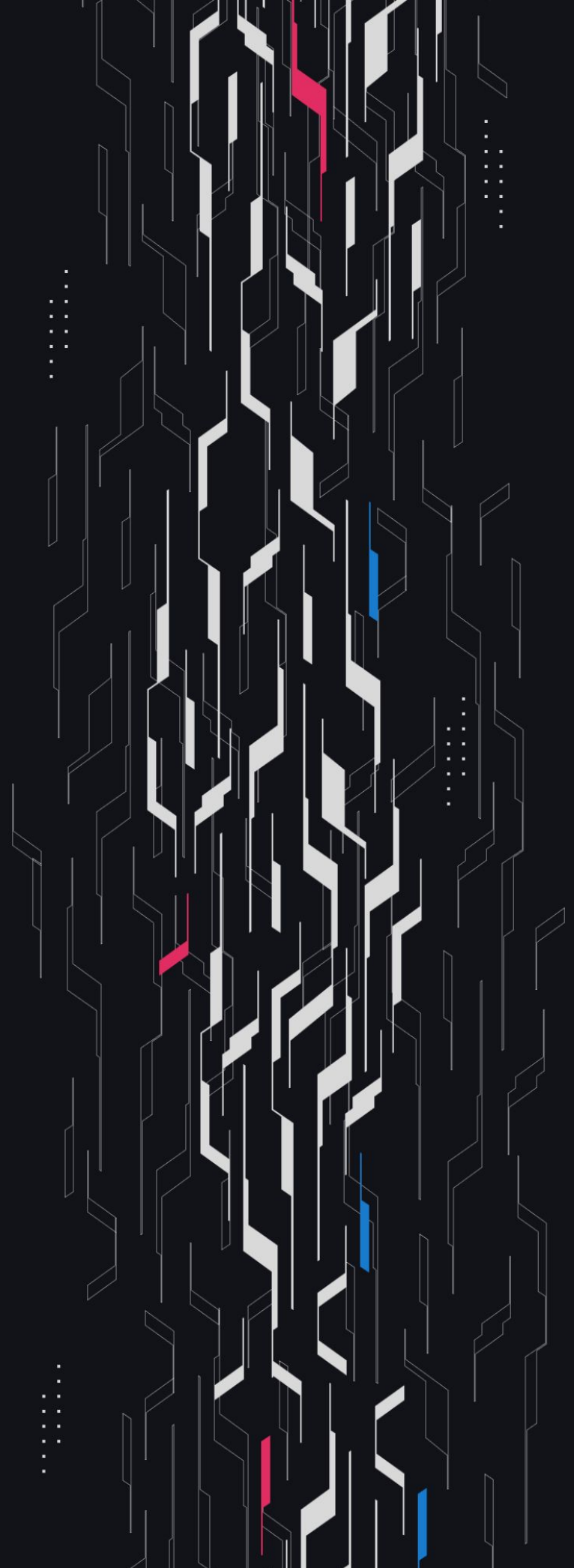
GA GUARDIAN

USDT0

**Conflux Deployment
and Peer Verification**

Security Assessment

November 12th, 2025



Summary

Audit Firm Guardian

Prepared By Daniel Gelfand

Client Firm USDT0

Final Report Date November 12, 2025

Audit Summary

USDT0 engaged Guardian to review the security of their USDT0 and CNHT0 Conflux Deployment and Peer Verification. From the 30th of October to the 31st of October, an auditor reviewed the source code in scope. All findings have been recorded in the following report.

Confidence Ranking

Given the lack of critical issues detected and minimal code changes following the main review, Guardian assigns a Confidence Ranking of 5 to the protocol. Guardian advises the protocol to consider periodic review with future changes. For detailed understanding of the Guardian Confidence Ranking, please see the rubric on the following page.

✓ Verify the authenticity of this report on Guardian's GitHub: <https://github.com/guardianaudits>

Guardian Confidence Ranking

Confidence Ranking	Definition and Recommendation	Risk Profile
5: Very High Confidence	Codebase is mature, clean, and secure. No High or Critical vulnerabilities were found. Follows modern best practices with high test coverage and thoughtful design. Recommendation: Code is highly secure at time of audit. Low risk of latent critical issues.	0 High/Critical findings and few Low/Medium severity findings.
4: High Confidence	Code is clean, well-structured, and adheres to best practices. Only Low or Medium-severity issues were discovered. Design patterns are sound, and test coverage is reasonable. Small changes, such as modifying rounding logic, may introduce new vulnerabilities and should be carefully reviewed. Recommendation: Suitable for deployment after remediations; consider periodic review with changes.	0 High/Critical findings. Varied Low/Medium severity findings.
3: Moderate Confidence	Medium-severity and occasional High-severity issues found. Code is functional, but there are concerning areas (e.g., weak modularity, risky patterns). No critical design flaws, though some patterns could lead to issues in edge cases. Recommendation: Address issues thoroughly and consider a targeted follow-up audit depending on code changes.	1 High finding and ≥ 3 Medium. Varied Low severity findings.
2: Low Confidence	Code shows frequent emergence of Critical/High vulnerabilities ($\sim 2/\text{week}$). Audit revealed recurring anti-patterns, weak test coverage, or unclear logic. These characteristics suggest a high likelihood of latent issues. Recommendation: Post-audit development and a second audit cycle are strongly advised.	2-4 High/Critical findings per engagement week.
1: Very Low Confidence	Code has systemic issues. Multiple High/Critical findings ($\geq 5/\text{week}$), poor security posture, and design flaws that introduce compounding risks. Safety cannot be assured. Recommendation: Halt deployment and seek a comprehensive re-audit after substantial refactoring.	≥ 5 High/Critical findings and overall systemic flaws.

Table of Contents

Project Information

Project Overview 5

Audit Scope & Methodology 7

Smart Contract Risk Assessment

Findings & Resolutions 10

Addendum

Disclaimer 19

About Guardian 20

Project Overview

Project Summary

Project Name	USDT0
Language	Solidity
Codebase	https://github.com/Everdawn-Labs/usdt0-oft-contracts
Commit	Initial commit: 5cada802c8d6e3e1f3eea34ea73b7ec19613abe0
Addresses	USDT0-OFT - Implementation: 0x2346d88699182f7c15cbfaa6e8824995eafab67a USDT0-OFT - OFT: 0xC57efa1c7113D98BdA6F9f249471704Ece5dd84A USDT0-OFT - Proxy Admin: 0xa882c21C9df00958A958cde96f2B2Ae8FB4315B1 USDT0 - Implementation: 0xaf37E8B6C9ED7f6318979f56Fc287d76c30847ff USDT0 - Proxy: 0x9151434b16b9763660705744891fA906F660EcC5 USDT0 - ProxyAdmin: 0xcb768e263FB1C62214E7cab4AA8d036D76dc59CC CNHT-OFT - OFT (Conflux): 0x0200C29006150606B650577BBE7B6248F58470c1 CNHT-OFT - Implementation (Conflux): 0x5be536efB28c8EfbD2da6894996f5df88347d8B2 CNHT-OFT - ProxyAdmin (Conflux): 0xa882c21C9df00958A958cde96f2B2Ae8FB4315B1 CNHT-OFT - OFT (Ethereum) : 0xA3CDdE086701839FbE255076d4730fF33F86C1cE CNHT-OFT - Implementation (Ethereum): 0x042dD806c8C8f17Fa89045fC5c79335eDb079772 CNHT-OFT - ProxyAdmin (Ethereum): 0x63eefF1b216f18A16a7499F20b6e6708634ca458 CNHT - Proxy (Conflux): 0xDEd1660192d4d82e7c0B628ba556861EdBB5CAda CHNT - Implementation (Conflux): 0xedaba024be4d87974d5aB11C6Dd586963CcCB027 CHNT - ProxyAdmin (Conflux): 0x21cAef8A43163Eea865baeE23b9C2E327696A3bf CNHT Token (Ethereum): 0x6e109e9dd7fa1a58bc3eff667e8e41fc3cc07aef

Project Overview

Audit Summary

Delivery Date	November 12, 2025
Audit Methodology	Static Analysis, Manual Review, Test Suite, Contract Fuzzing

Vulnerability Summary

Vulnerability Level	Total	Pending	Declined	Acknowledged	Partially Resolved	Resolved
● Critical	0	0	0	0	0	0
● High	0	0	0	0	0	0
● Medium	1	0	0	0	0	1
● Low	4	0	0	3	0	1
● Info	2	0	0	0	0	2

Audit Scope & Methodology

Addresses:

USDT0-OFT - Implementation: 0x2346d88699182f7c15cbfaa6e8824995eafab67a

USDT0-OFT - OFT: 0xC57efa1c7113D98BdA6F9f249471704Ece5dd84A

USDT0-OFT - Proxy Admin: 0xa882c21C9df00958A958cde96f2B2Ae8FB4315B1

USDT0 - Implementation: 0xaf37E8B6C9ED7f6318979f56Fc287d76c30847ff

USDT0 - Proxy: 0x9151434b16b9763660705744891fA906F660EcC5

USDT0 - ProxyAdmin: 0xcb768e263FB1C62214E7cab4AA8d036D76dc59CC

CNHT-OFT - OFT (Conflux): 0x0200C29006150606B650577BBE7B6248F58470c1

CNHT-OFT - Implementation (Conflux): 0x5be536efB28c8EfbD2da6894996f5df88347d8B2

CNHT-OFT - ProxyAdmin (Conflux): 0xa882c21C9df00958A958cde96f2B2Ae8FB4315B1

CNHT-OFT - OFT (Ethereum) : 0xA3CDdE086701839FbE255076d4730fF33F86C1cE

CNHT-OFT - Implementation (Ethereum): 0x042dD806c8C8f17Fa89045fC5c79335eDb079772

CNHT-OFT - ProxyAdmin (Ethereum): 0x63eefF1b216f18A16a7499F20b6e6708634ca458

CNHT - Proxy (Conflux): 0xDEd1660192d4d82e7c0B628ba556861EdBB5CAda

CHNT - Implementation (Conflux): 0xedaba024be4d87974d5aB11C6Dd586963CcCB027

CHNT - ProxyAdmin (Conflux): 0x21cAef8A43163Eea865baeE23b9C2E327696A3bf

CNHT Token (Ethereum): 0x6e109e9dd7fa1a58bc3eff667e8e41fc3cc07aef

USDT0 on Conflux:

<https://github.com/Everdawn-Labs/usdt0-tether-contracts-hardhat/pull/82/files>

<https://github.com/Everdawn-Labs/usdt0-oft-contracts/pull/150/files>

<https://github.com/Everdawn-Labs/usdt0-oft-contracts/pull/152/files>

<https://github.com/Everdawn-Labs/usdt0-oft-contracts/tree/conflux-monad-deployment/deployments/c\nflux-mainnet>

Audit Scope & Methodology

Vulnerability Classifications

Severity	Impact: <i>High</i>	Impact: <i>Medium</i>	Impact: <i>Low</i>
Likelihood: <i>High</i>	● Critical	● High	● Medium
Likelihood: <i>Medium</i>	● High	● Medium	● Low
Likelihood: <i>Low</i>	● Medium	● Low	● Low

Impact

- High** Significant loss of assets in the protocol, significant harm to a group of users, or a core functionality of the protocol is disrupted.
- Medium** A small amount of funds can be lost or ancillary functionality of the protocol is affected. The user or protocol may experience reduced or delayed receipt of intended funds.
- Low** Can lead to any unexpected behavior with some of the protocol's functionalities that is notable but does not meet the criteria for a higher severity.

Likelihood

- High** The attack is possible with reasonable assumptions that mimic on-chain conditions, and the cost of the attack is relatively low compared to the amount gained or the disruption to the protocol.
- Medium** An attack vector that is only possible in uncommon cases or requires a large amount of capital to exercise relative to the amount gained or the disruption to the protocol.
- Low** Unlikely to ever occur in production.

Audit Scope & Methodology

Methodology

Guardian is the ultimate standard for Smart Contract security. An engagement with Guardian entails the following:

- Two competing teams of Guardian security researchers performing an independent review.
- A dedicated fuzzing engineer to construct a comprehensive stateful fuzzing suite for the project.
- An engagement lead security researcher coordinating the 2 teams, performing their own analysis, relaying findings to the client, and orchestrating the testing/verification efforts.

The auditing process pays special attention to the following considerations:

- Testing the smart contracts against both common and uncommon attack vectors.
- Assessing the codebase to ensure compliance with current best practices and industry standards.
- Ensuring contract logic meets the specifications and intentions of the client.
- Cross-referencing contract structure and implementation against similar smart contracts produced by industry leaders.
- Thorough line-by-line manual review of the entire codebase by industry experts.
Comprehensive written tests as a part of a code coverage testing suite.
- Contract fuzzing for increased attack resilience.

Findings & Resolutions

ID	Title	Category	Severity	Status
M-01	CHNT0 LZReceive Cannot Credit On Mainnet	Warning	● Medium	Acknowledged
L-01	Delegate Not Multi-Sig	Warning	● Low	Resolved
L-02	Multiple Chains Not Connected To Conflux	Warning	● Low	Acknowledged
L-03	Duplicate Nonces On Safe	Warning	● Low	Acknowledged
I-01	Conflux Safe Has Different Signers	Warning	● Info	Resolved
I-02	Peers/Configs Not Set	Warning	● Info	Resolved

M-01 | CHNT0 LZReceive Cannot Credit On Mainnet

Category	Severity	Location	Status
Warning	● Medium		Resolved

Description

The CHNT token on mainnet returns a false boolean value upon transfer since `StandardTokenWithFees.transfer` does not explicitly return a bool, defaulting to false:

```
function transfer(address _to, uint _value) public returns (bool) {  
    uint fee = calcFee(_value);  
    uint sendAmount = _value.sub(fee);  
    super.transfer(_to, sendAmount);  
    if (fee > 0) {  
        super.transfer(owner(), fee);  
    }  
}
```

Consequently, whenever an `IzReceive` message is received by the OFT on mainnet, any attempt to transfer will fail due to `SafeERC20` reverting with `SafeERC20FailedOperation`, as the `false` return is interpreted as a failed transfer.

[illegible]

Recommendation

Adjust the `_credit` and `_debit` functions of the OFT Adapter on mainnet to avoid the `safeTransfer`.

Resolution

USDT0 Team: Resolved in the following PR:

<https://github.com/Everdawn-Labs/usdt0-oft-contracts/pull/161>.

L-01 | Delegate Not Multi-Sig

Category	Severity	Location	Status
Warning	● Low	Global	Resolved

Description

The delegate for CHNT0 OFT 0x0200C29006150606B650577BBE7B6248F58470c1 on Conflux is 0x1a6362AD64ccFF5902D46D875B36e8798267d154 which is not a multi-sig nor does it match the Safe address of 0x789498616f3Bb9F8dF52288A9311247028872105.

Recommendation

Set the delegate to 0x789498616f3Bb9F8dF52288A9311247028872105

Resolution

USDT0 Team: Resolved.

L-02 | Multiple Chains Not Connected To Conflux

Category	Severity	Location	Status
Warning	Low		Acknowledged

Description

Chains Flare and Monad are not wired for the latest USDT0 deployment on Conflux.

Recommendation

Ensure this is expected behavior.

Resolution

USDT0 Team: Acknowledged.

L-03 | Duplicate Nonces On Safe

Category	Severity	Location	Status
Warning	● Low		Acknowledged

Description

Both Conflux and Mantle Safes have duplicate nonce 40, hence the second transaction to `setReceiveLibrary` will revert with `LZ_SameValue()`. This is due to the Safe API's faulty pagination where the item count exceeds the limit on a page, hence a nonce can appear twice and items between pages are not necessarily unique.

40	<> Contract interaction	1 day ago	1 out of 3	Confirm	▼
40	<> Contract interaction	1 day ago	1 out of 3	Confirm	▼

Recommendation

Ensure not to execute a duplicate and notify Safe API of this pagination issue.

Resolution

USDT0 Team: Acknowledged.

I-01 | Conflux Safe Has Different Signers

Category	Severity	Location	Status
Warning	● Info	Global	Resolved

Description

Celo Safe has signers:

- 1. 0x4192D72c281dB0eF10464dD274EE583C33256ac5
- 2. 0x00F6D2b4B69Ce697f913Da16A9D73283dc4C78F2
- 3. 0x488A3d37442F583c1c0a356118d1d6181b22434c
- 4. 0x7D8e979dcC6EC933c70FBe28B2B7700907D2aFEA
- 5. 0x5F0DcA105195Bd58277aa4b4CABC90a791E1b982

But the Conflux Safe at 0x789498616f3Bb9F8dF52288A9311247028872105 has signers:

- 1. 0x1a6362AD64ccFF5902D46D875B36e8798267d154
- 2. 0x7D8e979dcC6EC933c70FBe28B2B7700907D2aFEA
- 3. 0x488A3d37442F583c1c0a356118d1d6181b22434c
- 4. 0x4192D72c281dB0eF10464dD274EE583C33256ac5
- 5. 0x5F0DcA105195Bd58277aa4b4CABC90a791E1b982

0x00F6D2b4B69Ce697f913Da16A9D73283dc4C78F2 is missing from the Conflux Safe and 0x1a6362AD64ccFF5902D46D875B36e8798267d154 was added.

Recommendation

Ensure the signers are intended on the Conflux Safe.

Resolution

USDT0 Team: Resolved.

I-02 | Peers/Configs Not Set

Category	Severity	Location	Status
Warning	● Info	Global	Resolved

Description

Peers and configs are not currently set for Conflux and other chains, hence the existing block confirmations/DVN's/etc. do not currently match the intended configs.

Recommendation

Ensure intended configs are set before cross-chain transfer is allowed.

Resolution

USDT0 Team: Resolved.

Remediation Findings & Resolutions

ID	Title	Category	Severity	Status
L-01	CNHT Token Does Not Support Zero Address	Warning	Low	Acknowledged

L-01 | CNHT Token Does Not Support Zero Address

Category	Severity	Location	Status
Warning	● Low		Acknowledged

Description

Unlike the USDT token, the CNHT token reverts on zero address transfer. Hence debited funds on src chain will not be credited on mainnet and the LayerZero message will be unfulfilled.

Recommendation

Clearly document this behavior and consider crediting address(0) funds to the 0xdead address.

Resolution

USDT0 Team: Acknowledged.

Disclaimer

This report is not, nor should be considered, an “endorsement” or “disapproval” of any particular project or team. This report is not, nor should be considered, an indication of the economics or value of any “product” or “asset” created by any team or project that contracts Guardian to perform a security assessment. This report does not provide any warranty or guarantee regarding the absolute bug-free nature of the technology analyzed, nor do they provide any indication of the technologies proprietors, business, business model or legal compliance.

This report should not be used in any way to make decisions around investment or involvement with any particular project. This report in no way provides investment advice, nor should be leveraged as investment advice of any sort. This report represents an extensive assessing process intending to help our customers increase the quality of their code while reducing the high level of risk presented by cryptographic tokens and blockchain technology.

Blockchain technology and cryptographic assets present a high level of ongoing risk. Guardian’s position is that each company and individual are responsible for their own due diligence and continuous security. Guardian’s goal is to help reduce the attack vectors and the high level of variance associated with utilizing new and consistently changing technologies, and in no way claims any guarantee of security or functionality of the technology we agree to analyze.

The assessment services provided by Guardian is subject to dependencies and under continuing development. You agree that your access and/or use, including but not limited to any services, reports, and materials, will be at your sole risk on an as-is, where-is, and as-available basis. Cryptographic tokens are emergent technologies and carry with them high levels of technical risk and uncertainty. The assessment reports could include false positives, false negatives, and other unpredictable results. The services may access, and depend upon, multiple layers of third-parties.

Notice that smart contracts deployed on the blockchain are not resistant from internal/external exploit. Notice that active smart contract owner privileges constitute an elevated impact to any smart contract’s safety and security. Therefore, Guardian does not guarantee the explicit security of the audited smart contract, regardless of the verdict.

About Guardian

Founded in 2022 by DeFi experts, Guardian is a leading audit firm in the DeFi smart contract space. With every audit report, Guardian upholds best-in-class security while achieving our mission to relentlessly secure DeFi.

To learn more, visit <https://guardianaudits.com>

To view our audit portfolio, visit <https://github.com/guardianaudits>

To book an audit, message <https://t.me/guardianaudits>